METRIC
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10 January 2000
SUPERSEDING
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PERFORMANCE SPECIFICATION

CLEANING COMPOUND, DECONTAMINATING (FOR SOILED AND RADIOACTIVE CONTAMINATED SURFACES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for a decontaminating cleaning compound, for soiled and radioactive contaminated surfaces, that is soluble in water, hard water and sea water (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6850

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issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-2904	Thinner, Paint, Mineral Spirits, Regular and Odorless
QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-N-95	Naphtha; Aliphatic

DEPARTMENT OF DEFENSE

MIL-PRF-5425	Plastic Sheet, Acrylic, Heat Resistant
MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-C-8514	Coating Compound, Metal Pretreatment, Resin-Acid (ASG)
MIL-PRF-22750	Coating, Epoxy, High Solids
MIL-PRF-23377	Primer Coatings; Epoxy, High Solid
MIL-PRF-81352	Coating, Aircraft, Touch-Up

(Unless otherwise indicated, copies of the above specifications are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 187	Copper Bar, Bus Bar, Rod and Shapes, Standard Specification for (DoD Adopted)
ASTM B 272	Copper Flat Products with Finished (Rolled or Drawn) Edges (Flat Wire and Strip), Standard Specification for (DoD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS-QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet (DoD Adopted)
AMS-QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Sheet (DoD Adopted)
AMS M3171	Magnesium Alloy, Process for Pretreatment and Prevention of Corrosion on (DoD Adopted)
AMS 4375	Sheet and Plate, Magnesium Alloy 3.0Al - 1.0Zn - 0.20Mn (AZ31B-0) Annealed and Recrystallized (DoD Adopted)
AMS 4376	Plate, Magnesium Alloy 3.0Al - 1.0Zn - 0.20Mn (AZ31B-H26) Cold Rolled and Partially Annealed (DoD Adopted)
AMS 4377	Sheet and Plate, Magnesium Alloy 3.0Al - 1.0Zn - 0.20Mn (AZ31B-H24) Cold Rolled, Partially Annealed (DoD Adopted)
AMS 5510	Steel, Corrosion and Heat Resistant, Sheet, Strip and Plate 18Cr - 10.5Ni - 0.40Ti (SAE 30321) Solution Heat Treated (DoD Adopted)
AMS 5512	Steel, Corrosion and Heat Resistant, Sheet, Strip, and Plate 18Cr - 10.5Ni - 0.80Cb (SAE 30347) Solution Heat Treated (DoD Adopted)
SAE J 1966	Oil, Lubricating, Aircraft Piston Engine (Nondispersant Mineral Oil) (DoD Adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, Inc. 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Physical form. When tested as specified in 4.3.1, the raw materials used in the manufacture of this cleaning compound shall be capable of being intimately assembled and processed to produce a homogeneous mixture free of inhomogeneity, lumps and foreign material.

3.2 Solubility in distilled water. When tested as specified in 4.3.2, the cleaning compound shall dissolve readily in distilled water producing no more than a slight cloudiness. If the extent of the cloudiness is questionable, the volume of sediment, after centrifuging, shall be not more than 0.05 percent.

3.3 Stability in distilled water. After being subjected to temperatures of 4 ± 1 °C and 98 ± 1 °C for 10-minute periods, as specified in 4.3.3, the cleaning compound-distilled water

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solution shall show no sign of separation, sedimentation, gelling or noticeable change in viscosity and only a slight cloudiness. If the extent of the cloudiness is questionable, the volume of sediment, after centrifuging, shall be not more than 0.05 percent.

3.4 Solubility in synthetic hard water. When tested as specified in 4.3.4, the volume of sediment in the cleaning compound-synthetic hard water solution shall be not more than 0.8 percent.

3.5 Solubility in synthetic sea water. When tested as specified in 4.3.5, the volume of sediment in the cleaning compound-synthetic sea water solution shall be not more than 7.0 percent.

3.6 pH. When tested as specified in 4.3.6, the pH of the cleaning compound-distilled water solution shall be not less than 8.0 nor more than 10.0.

3.7 Total alkalinity. When tested as specified in 4.3.7, the total alkalinity of the cleaning compound, calculated as Na_2O, shall be not more than 21 percent.

3.8 Surface tension. When tested as specified in 4.3.8, the surface tension of the cleaning compound test solutions shall conform to the requirements of table I.

TABLE I. Surface tension.

Cleaning compound dissolved in	Surface tension, maximum (in Newtons per meter)
Distilled water	3.4×10^{-2}
Synthetic hard water	3.2×10^{-2}
Synthetic sea water	3.2×10^{-2}

3.9 Interfacial tension. When tested as specified in 4.3.9, the interfacial tension between a layer of white paraffin oil and the specified cleaning compound test solution shall conform to the requirements of table II.

TABLE II. Interfacial tension.

Cleaning compound dissolved in	Interfacial tension, maximum (in Newtons per meter)
Distilled water	4×10^{-3}
Synthetic hard water	4×10^{-3}
Synthetic sea water	3×10^{-3}

3.10 Radioactive soil removal. When tested as specified in 4.3.10, the cleaning compound-distilled water solution shall remove not less than 20 percent of the radioactivity from the soil retaining plugs.

3.11 Cleaning ability. When tested as specified in 4.3.11, the cleaning compound test solutions shall completely remove an oil-carbon mixture from the test panels.

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3.12 Rinsing ability. When tested as specified in 4.3.12, the cleaning compound shall be completely removed from the test panels after rinsing with distilled water, synthetic hard water or synthetic sea water; except that after rinsing with synthetic hard water a slight film on the panels shall be acceptable.

3.13 Corrosiveness. When tested as specified in 4.3.13, the cleaning compound-distilled water solution shall cause no discoloration, corrosion, chemical attack or any other deterioration of the test panels; except that a slight streaking of the aluminum alloy panel and a slight dulling of the alclad aluminum alloy panel shall be acceptable.

3.14 Effect on painted surfaces. When tested as specified in 4.3.14, the cleaning compound test solutions shall cause no softening, blistering, discoloration, cracking or any other deterioration of the epoxy-coated or lacquered test panels.

3.15 Effect on acrylic-base plastics. When tested as specified in 4.3.15, the cleaning compound test solutions shall cause no crazing, chemical attack, or any other deterioration of the stressed acrylic panels.

4. VERIFICATION

4.1 Inspection conditions. Unless otherwise specified herein, all tests shall be performed at a laboratory temperature of 23 ± 1°C and a relative humidity of 50 ± 4 percent.

4.2 Conformance inspection. The characteristics shown in 3.1 through 3.15, when tested in accordance with 4.3, shall constitute minimum inspections to be performed by the supplier prior to Government acceptance or rejection. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies, submitted to the Government for acceptance, comply with all requirements of the contract.

4.2.1 Sampling for conformance inspection. For the purpose of sampling, a lot shall consist of a manufacturer's batch. If the material cannot be identified by batch, a lot shall consist of not more than 4,500 kg of cleaning compound. A batch is defined as the end product of all raw materials mixed or blended in a single operation.

4.2.2 Samples for test. Two 2-kg samples of cleaning compound shall be selected at random from each lot, either during or after the filling operation.

4.3 Test methods.

4.3.1 Physical form. When visually inspected, the raw materials used in the manufacture of the cleaning compound shall conform to the requirements of 3.1.

4.3.2 Solubility in distilled water. Add 43 g of the cleaning compound to 6 L of distilled water and mix thoroughly. After allowing to stand undisturbed for 5 minutes, observe the solution for

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cloudiness. If the extent of cloudiness is questionable, the volume of the sediment shall be determined as specified in 4.3.2.1.

4.3.2.1 Sediment. One hundred mL of the cleaning compound test solution shall be added to each of two 100 mL graduated centrifuge tubes, graduated for reading to 0.1 mL. Then the tubes shall be centrifuged at a speed of 1500-1800 rpm for 15-minute periods until the volume of sediment in each tube becomes constant. The sediment volumes shall be read to 0.1 mL and the average of the two results shall be reported. The diameter of swing (tip to tip of the whirling tubes) of the centrifuge shall be between 40.6 and 45.7 cm.

4.3.3 Stability in distilled water. Add 50 mL of the cleaning compound-distilled water solution specified in 4.3.2 to each of two test tubes. Cool one tube to 4 ± 1°C and heat the other to 98 ± 1°C. After 10 minutes at these temperatures, condition at room temperature for 18 hours and then observe each tube for any evidence of separation, sedimentation, gelling, noticeable change in viscosity or cloudiness. If the extent of cloudiness is questionable, the volume of sediment shall be determined as specified in 4.3.2.1.

4.3.4 Solubility in synthetic hard water. Prepare synthetic hard water by dissolving 3.0 g of calcium acetate monohydrate, and 2.1 g of magnesium sulfate heptahydrate, in 8 L of distilled water. Add 43 g of the cleaning compound to 6 L of the synthetic hard water and mix thoroughly. After allowing to remain undisturbed for 5 minutes, determine the volume of sediment as specified in 4.3.2.1.

4.3.5 Solubility in synthetic sea water. The synthetic sea water shall be prepared as specified in table III. Add 43 g of the cleaning compound to 6 L of the synthetic sea water and mix thoroughly. After allowing to remain undisturbed for 5 minutes, determine the volume of sediment as specified in 4.3.2.1.

4.3.6 pH. The cleaning compound-distilled water test solution specified in 4.3.2 shall be tested with a pH electrometer having a sensitivity and readability of 0.05.

4.3.7 Total alkalinity.

4.3.7.1 Indicator solution. Grind a 0.1 g sample of bromophenol blue in a mortar with 1.5 mL of 0.1 N sodium hydroxide. Transfer this mixture to a volumetric flask, dilute to 250 mL with distilled water, and adjust to neutrality.

TABLE III. Synthetic sea water.

Ingredient	Quantity in grams
Sodium chloride	150.0
Magnesium chloride, hexahydrate	66.0
Calcium chloride, dihydrate	9.6
Sodium sulfate, anhydrous	24.0
Add distilled water to make 6 L of solution	

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4.3.7.2 Procedure. Add 5 drops of the indicator solution prepared in 4.3.7.1 to a 100 mL aliquot of the cleaning compound-distilled water solution specified in 4.3.2. A blue coloration should develop. Titrate the solution with 0.5 N hydrochloric or sulfuric acid to the first appearance of yellow. Calculate the total alkalinity as follows:

$$\text{Percent total alkalinity (as } Na_2O) = \frac{3.1\ (A)\ (N)}{0.720}$$

where A = mL of acid
N = normality of acid

4.3.8 Surface tension.

4.3.8.1 Equipment. Surface tension measurements shall be made with an instrument capable of reading to 1×10^{-4} N/m with a precision of $\pm 5 \times 10^{-5}$ N/m.

4.3.8.2 Measurements. Surface tension measurements shall be made on a 50 mL portion of each of the cleaning compound test solutions specified in 4.3.2, 4.3.4 and 4.3.5.

4.3.9 Interfacial tension. Using the apparatus specified in 4.3.8.1, interfacial tension measurements shall be made after a portion of each of the cleaning compound test solutions specified in 4.3.2, 4.3.4 and 4.3.5 have been covered with an approximately equal volume of white paraffin oil.

4.3.10 Radioactive soil removal.

4.3.10.1 Radioactive soil. A partial formulation of the radioactive soil shall be made as specified in table IV. Heat the mixture in a porcelain crucible at a temperature of 1000 to 1200°C for 3 hours to convert the ingredient metals to their oxides. Cool to room temperature and transfer to a stainless steel ball mill that is one-half filled with stainless steel balls. Add 50 g of uranium oxide and 70 mL of distilled water to the mill. (Uranium oxide can be prepared by heating uranium acetate at 1000 to 1200°C in an electric furnace.) After letting the mill run for 3 hours, transfer the mixture to a beaker. Rinse the mill and balls with distilled water, adding the washings to the beaker. Continue rinsing the mill until the washings, after evaporation, exhibit no appreciable radiation as measured by a Geiger-Mueller Counter. Reduce the volume of the soil slurry to 1 L by evaporation. Add 50 g of uranyl nitrate to the slurry and mix thoroughly.

4.3.10.2 Soil retaining plugs. Soil retaining plugs shall be constructed from Alclad 2024 aluminum alloy and be a 2.5 cm ring and disc assembly. The plugs shall be cleaned, chemically treated and coated with the lacquer system specified in 4.3.14.1.

4.3.10.3 Application of radioactive soil. Place a tapered stainless steel cylinder over each of five soil retaining plugs prepared as specified in 4.3.10.2. The radioactive soil slurry prepared in 4.3.2 shall be then shaken vigorously and a small portion of it transferred to a graduated cylinder fitted with a ground glass stopper. Dilute the slurry with 4 times its volume of distilled water. After shaking vigorously, transfer 0.5 mL of the diluted slurry to each plug by means of a pipet.

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Evaporate the water from each plug using an infra-red lamp under a hood. Remove the cylinder from the plugs after the major portion of the water has been evaporated and continue heating until the plugs are dry.

TABLE IV. Radioactive soil (partial formulation).

Ingredients	Weight in grams
Lanthanum nitrate	16.93
Molybdenum oxide	16.40
Ferric oxide	11.11
Nickel chloride	0.08
Mercuric oxide	0.03
Stannous chloride	0.38
Tellurium dioxide	7.41
Antimony trioxide	36.48
Cerium oxide	10.69
Arsenic oxide	0.33
Cobalt nitrate	0.08
Lead oxide	0.08
Zinc oxide	Trace
Selenious acid	Trace

4.3.10.4 Procedure. Place the five test plugs containing the dry radioactive soil (that were prepared in 4.3.10.3) in a liter beaker with 700 mL of distilled water. Position a glass rod attached to an electric stirrer in the beaker so that its end is 2.5 cm above the plugs. Stir the contents of the beaker for 5 minutes with the speed of the stirrer adjusted so that the plugs do not rotate. Remove the plugs, wash off in a strong stream of distilled water from a wash bottle, and dry in a fume hood under an infra-red heat lamp. Radioactivity measurements (R_a) shall then be made on each plug using an end window tube Geiger-Mueller Counter. After the measurements have been taken, place the five plugs in 1 L beaker with 700 mL of the cleaning compound-distilled water solution specified in 4.3.2. Arrange the stirrer as described above and stir for 30 minutes. Remove the plugs and rinse in a strong stream of distilled water from a wash bottle. Dry in a fume hood under an infra-red heat lamp. Radioactivity measurements (R_b) shall then be taken on each plug. Calculations shall be made as follows:

$$\text{Percent radioactivity removal} = \frac{(R_a - R_b) \times 100}{R_a}$$

where R_a= Radiation count, corrected for background, after cleaning with distilled water

R_b= Radiation count, corrected for background, after cleaning with the cleaning compound-distilled water solution

Note: The radiation count shall be recorded as counts per second

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4.3.11 Cleaning ability. An epoxy-coated and a lacquered panel shall be prepared as specified in 4.3.14.1. Observe each panel under an ultraviolet light for fluorescence. Replace any panel that shows any sign of fluorescence. Spread 2 drops of oil, conforming to SAE J 1966, evenly over each panel. Then sift carbon black through a cheesecloth and spread evenly over each panel. Turn the panels over and tap lightly to remove excess carbon. Pour 10 mL of the cleaning compound-distilled water solution specified 4.3.2 onto each panel and spread evenly over the soiled surface. Place each panel in a horizontal position in a tray. Clean each panel by applying another 10 mL portion of the cleaning compound-distilled water solution and rubbing with absorbent cotton. Flood the panels once again with cleaning compound-distilled water solution and rinse thoroughly with distilled water. Allow to dry at laboratory room temperature at an angle of 45 degrees from the vertical. Observe the dried panels for the presence of oil as evidenced by iridescence or any increase in the gloss of the panels. Observe the panels under the ultraviolet light for fluorescence resulting from any residual oil on the panels. Repeat this test using the cleaning compound-synthetic hard water solution specified in 4.3.4 with synthetic hard water rinse and the cleaning compound-synthetic sea water solution specified in 4.3.5 with synthetic sea water rinse.

4.3.12 Rinsing ability. An epoxy-coated and a lacquered panel shall be prepared as specified in 4.3.14.1. Flood each panel with the cleaning compound-distilled water solution specified in 4.3.2 and allow to drain and air dry at an angle of 45 degrees from the vertical. Then place a 250 Watt infra-red lamp 30 cm from the panels. After exposing the panels to the heat of the lamp for 1 hour, rinse the panels thoroughly with distilled water. Observe for any indication of a cleaning compound residue remaining on the panels. Repeat this test using the cleaning compound-synthetic hard water solution specified in 4.3.4 and the cleaning compound-synthetic sea water solution specified in 4.3.5, rinsing respectively with synthetic hard water and synthetic sea water.

4.3.13 Corrosiveness.

4.3.13.1 Test panels. Test panels measuring 0.127 by 2.5 by 15 cm shall be prepared as specified in table V.

4.3.13.2 Procedure. Immerse the six test panels in the cleaning compound-distilled water solution specified in 4.3.2. Do not allow the panels to contact each other. After a 1-hour immersion, remove the panels and allow to remain flat in a horizontal position for 24 hours. After rinsing thoroughly with distilled water, examine each panel for discoloration, corrosion, chemical attack or any other sign of deterioration.

4.3.14 Effect on painted surfaces. An epoxy-coated and a lacquered panel shall be prepared as specified in 4.3.14.1. Immerse both panels in the cleaning compound-distilled water solution specified in 4.3.2. After 45 minute immersion, remove the panels and allow to drain for 5 minutes at an angle of 45 degrees from the vertical. Rinse the panels thoroughly with distilled water and examine the painted surface for softening, blistering, discoloration, cracking or any other sign of deterioration. Repeat this test, immersing a similar set of panels in the cleaning compound-synthetic hard water solution specified in 4.3.4 and in the cleaning compound-synthetic sea water solution specified in 4.3.5.

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TABLE V. Test panels.

Panel	Specification
Aluminum alloy 2024, polished[1]	AMS-QQ-A-250/4
Aluminum alloy, Alclad 2024, polished[1]	AMS-QQ-A-250/5
Copper, polished	ASTM B 187 ASTM B 272
Steel, polished	AMS 5510 AMS 5512
Magnesium alloy[2]	AMS 4375 AMS 4376 AMS 4377
Steel[3]	AMS 5510

[1]Panels shall be polished by means of a buffing wheel and buffing compound.
[2]Treated in accordance with AMS M3171.
[3]Cadmium plated in accordance with QQ-P-416, Type I.

4.3.14.1 Painted test panels. Clean 7.6 by 15 cm panels conforming to aluminum alloy 2024, by brushing on a 50/50 mixture of an aluminum cleaning compound and mineral spirits conforming to A-A-2904, Type I. Allow the mixture to react with the panels for 15 to 30 minutes and then rinse thoroughly with distilled water. After the panels have dried, apply by brush or dip one coat of chemical film conforming to MIL-C-5541. After the chemical film has air dried for 24 hours, spray with one coat of wash primer conforming to MIL-C-8514 and air dry 45 minutes to a dry film thickness of 5.1 to 7.6 µm. The panels shall then be sprayed with either the epoxy or lacquer system as specified in table VI or VII.

TABLE VI. Coating - epoxy system.

Coating	Specification	Air dry time	Dry film thickness (in µm)
Primer	MIL-PRF-23377	1 hour	12.7 - 20.3
White epoxy	MIL-PRF-22750	1 hour	A thin mist coat
White epoxy	MIL-PRF-22750	7 hours between coats; final dry 8 days	35.6 - 45.7

TABLE VII. Coating - lacquer system.

Coating	Specification	Air dry time	Dry film thickness (in µm)
Primer		45 minutes	7.5-10.2
Sea blue lacquer	MIL-PRF-81352	45 minutes between coats; final dry of 24 hours	20.3-30.5

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4.3.15 Effect on acrylic-base plastics. Acrylic plastic conforming to MIL-PRF-5425 shall be cut into 2.5 by 17.8 by 0.64 cm test specimens. Clean the plastic strips with aliphatic naphtha conforming to TT-N-95 and condition at room temperature for at least 96 hours prior to use. Set up four plastic strips as Class 1 levers with the fulcrum 5 cm from the clamped end with a load of 2.4 kg suspended 10 cm from the fulcrum. This load produces an outer fiber stress of 13,780 kPa at the fulcrum. After 30 minutes, and while the strips remain under stress, examine for crazing as follows: Place a source of light under the strips and examine the fulcrum area with a 7-power magnifier at such an angle that light will be reflected from any cracks or fissures that may be present. Use the light source only during examination to prevent overheating of the plastic strips. Remove any strip that shows any sign of crazing. While the strips are under stress, place 10 drops of cleaning compound-distilled water solution specified in 4.3.2 at the fulcrum of each of three strips (the fourth is a control). Place a glass slide, 2.5 by 3.8 cm, over the liquid with its long axis parallel to that of the plastic strip. After the first application of test solution, apply an additional 2 drops of solution at intervals of 7, 24, and 31 hours to compensate for evaporation losses. Forty-eight hours after the first application of the test solution, and while the strips are under stress, examine for crazing as described above. All cracks at the edges of a strip shall be noted as "edge crazing" and disregarded unless they extend across the entire width of the strip. If the control crazes, repeat this test using four different test specimens. Examine the strips also for signs of chemical attack or any other deterioration. Repeat this test using the cleaning compound-synthetic hard water solution specified in 4.3.4 and the cleaning compound-synthetic sea water solution specified in 4.3.5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use.

6.1.1 Application. The cleaning compound, after dissolving in water, hard water or sea water, is intended for use in the cleaning of finished and unfinished aircraft surfaces that are soiled.

6.1.2 Military unique. The cleaning compound is specifically designed to be used in ten different weapon systems to clean military aircraft surfaces that are contaminated with radioactive material. No commercial equivalent exists.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3)
c. Unit and quantity required
d. Packaging requirements (see 5.1)

6.3 Composition. Table VIII is the suggested composition of the cleaning compound.

TABLE VIII. Composition.

Ingredient	Percent by weight
Sodium pyrophosphate (anhydrous)	10
Sodium tripolyphosphate (anhydrous)	30
Sodium phosphate, tribasic (anhydrous)	20
Non-ionic detergent	10
Methylcelluose (33-56 percent active ingredient)	5
Sodium xylenesulfonate	5
Sodium hexametaphosphate	10
Borax	10

6.4 Subject term (key word) listing.

Borax
Methylcelluose
Non-ionic detergent
Sodium hexametaphosphate
Sodium phosphate, tribasic
Sodium pyrophosphate
Sodium tripolyphosphate
Sodium xylenesulfonate

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy - AS

Preparing activity:
DLA - GS

(Project 6850-1214)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-7907B	**2. DOCUMENT DATE (YYYYMMDD)** 20000110

3. DOCUMENT TITLE **CLEANING COMPOUND, DECONTAMINATING (FOR SOILED AND RADIOACTIVE CONTAMINATED SURFACES)**

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER RICHMOND (DSCR) ATTN: DSCR-VBD	b. TELEPHONE *Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)* 8000 JEFFERSON DAVIS HIGHWAY RICHMOND, VA 23297-5610	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533 Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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